MECHATRONIC SYSTEM INTEGRATION
MCTA 3203

WEEK 2 :

DIGITAL LOGIC SYSTEM

SECTION 1

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Abstract

This experiment demonstrates the design and implementation of a manual decimal counter using an Arduino Uno and a common cathode 7-segment display. The project focuses on digital logic principles by utilizing pushbuttons for incrementing (0–9) and resetting the numerical output. Through the development of a switch-case program and the integration of current-limiting resistors, the system achieved accurate visual representation of decimal digits. The results confirmed a reliable interface between hardware and software, effectively managing state transitions while identifying mechanical bouncing as a primary source of error. This project validates fundamental mechatronic concepts in circuit control and binary-coded decimal application.

Introduction

This experiment aims to display numerical values from 0 to 9 using a 7-segment display controlled by an Arduino Uno. The objective is to demonstrate the working principles of simple digital logic circuits through programming.

The 7-segment display is commonly used in mechatronics projects to represent numerical outputs, where each segment is individually controlled to form digits. The circuit also incorporates current-limiting resistors, proper input-output pin configuration, and time delay functions within the Arduino program.

In this experiment, push buttons are used as inputs to control the displayed values. One button functions as an increment button to increase the displayed number from 0 to 9, while another button serves as a reset to return the display to 0. When the circuit is correctly wired and programmed, pressing the increment button updates the number accordingly on the display.

The successful operation of this setup verifies proper interfacing between the Arduino, the 7-segment display, and the push buttons, demonstrating the effective application of digital logic and circuit control principles.

Materials and Equipment

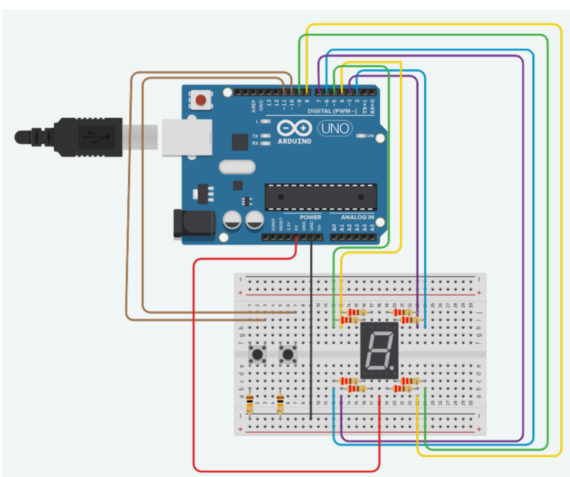
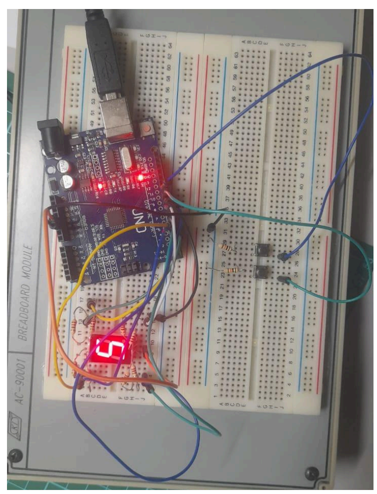
1. Arduino Uno board
2. Common cathode 7-segment display
3. 220-ohm resistors (8)
4. 10k-ohm resistors (2)
5. Pushbuttons (2)
6. Jumper wires
7. Breadboard

Experimental setup

Circuit Setup:

1. Connect the common cathode 7-segment display to the Arduino Uno as follows:
    - Connect each of the 7 segments (a, b, c, d, e, f, g) of the display to separate digital pins on the Arduino (e.g., D0 to D6).
    - Connect the common cathode pin of the display to one of the GND (ground) pins on the Arduino.
    - Use 220-ohm resistors to connect each of the segment pins to the Arduino pins to limit the current.

2. Connect the pushbuttons to the Arduino:
    - Connect one leg of each pushbutton to a separate digital pin (e.g., D9 and D10) and connect the other leg of each pushbutton to GND.
    - Use 10K-ohm pull-up resistors for each pushbutton by connecting one end of each resistor to the digital pin and the other end to the 5V output of the Arduino.

Methodology

Experiment steps:

1. Build the circuit according to the circuit setup instructions.
2. Upload the provided Arduino code to our Arduino Uno.
3. Open the Serial Monitor in the Arduino IDE.
4. Press the increment button to increase the count. The 7-segment display should show the numbers from 0 to 9 sequentially.
5. Press the reset button to reset the count to 0.

IDE code:

```
const int segmentA = 2;
const int segmentB = 3;
const int segmentC = 4;
const int segmentD = 5;
const int segmentE = 6;
const int segmentF = 7;
const int segmentG = 8;
const int INC_Button = 10;
const int RES_Button = 11;
bool lastIncButtonState= HIGH;
bool lastResButtonState= HIGH;
int i=0;
void setup() {
pinMode(segmentA, OUTPUT);
pinMode(segmentB, OUTPUT);
pinMode(segmentC, OUTPUT);
pinMode(segmentD, OUTPUT);
pinMode(segmentE, OUTPUT);
pinMode(segmentF, OUTPUT);
pinMode(segmentG, OUTPUT);
pinMode(INC_Button, INPUT_PULLUP);
pinMode(RES_Button, INPUT_PULLUP);
}
// 0 = A,B,C,D,E,F
// 1 = B,C
// 2 = A,B,G,E,D
// 3 = A,B,C,D,G
// 4 = A,F,G,C,D
void loop() {
switch (i)
{
  case 0: {
    digitalWrite(segmentA, LOW); digitalWrite(segmentB, LOW); digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, HIGH); digitalWrite(segmentE, LOW); digitalWrite(segmentF, LOW);
    digitalWrite(segmentG, LOW); break; }
  case 1: {
    digitalWrite(segmentA, LOW); digitalWrite(segmentB, HIGH); digitalWrite(segmentC, HIGH);
    digitalWrite(segmentD, HIGH); digitalWrite(segmentE, HIGH); digitalWrite(segmentF, HIGH);
```

```
  digitalWrite(segmentG, LOW); break; }
 case 2: {
  digitalWrite(segmentA, LOW); digitalWrite(segmentB, LOW); digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, LOW); digitalWrite(segmentE, LOW); digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, HIGH); break; }
 case 3: {
  digitalWrite(segmentA, LOW); digitalWrite(segmentB, LOW); digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, LOW); digitalWrite(segmentE, HIGH); digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, LOW); break;  }
 case 4: {
  digitalWrite(segmentA, LOW); digitalWrite(segmentB, HIGH); digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW); digitalWrite(segmentE, HIGH); digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, LOW); break; }
 case 5: {
  digitalWrite(segmentA, HIGH); digitalWrite(segmentB, LOW); digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW); digitalWrite(segmentE, HIGH); digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, LOW); break; }
 case 6: {
  digitalWrite(segmentA, HIGH); digitalWrite(segmentB, LOW); digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW); digitalWrite(segmentE, LOW); digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, LOW); break; }
 case 7: {
  digitalWrite(segmentA, LOW); digitalWrite(segmentB, LOW); digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH); digitalWrite(segmentE, HIGH); digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, LOW); break; }
 case 8: {
  digitalWrite(segmentA, LOW); digitalWrite(segmentB, LOW); digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW); digitalWrite(segmentE, LOW); digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, LOW); break; }
 case 9: {
  digitalWrite(segmentA, LOW); digitalWrite(segmentB, LOW); digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW); digitalWrite(segmentE, HIGH); digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, LOW); break; }
}
 bool IncButton=digitalRead(INC_Button);
 if(IncButton==LOW && lastIncButtonState== HIGH)
 {
  i++;
  if(i > 9)
```

```
    i = 0;
  delay(200);
}
  lastIncButtonState=IncButton;
  bool ResButton=digitalRead(RES_Button);
  if(ResButton==LOW && lastResButtonState== HIGH)
  {
    i = 0;
  delay(200);
}
  lastResButtonState=ResButton;
}
```

Data collection

No	Input	Observed Output
1	The arduino is turned on	The 7-segment displays ‘0’
2	The INC_Button is pressed	The 7-segment displays ‘1’
3	The INC_Button is pressed	The 7-segment displays ‘2’
4	The INC_Button is pressed	The 7-segment displays ‘3’
5	The INC_Button is pressed	The 7-segment displays ‘4’
6	The INC_Button is pressed	The 7-segment displays ‘5’
7	The INC_Button is pressed	The 7-segment displays ‘6’
8	The INC_Button is pressed	The 7-segment displays ‘7’
9	The INC_Button is pressed	The 7-segment displays ‘8’
10	The INC_Button is pressed	The 7-segment displays ‘9’
11	The INC_Button is pressed	The 7-segment displays ‘0’
12	The RES_Button is pressed	The 7-segment displays ‘0’
13	The RES_Button is pressed when the 7-segment display ‘7’	The 7-segment displays ‘0’

Data Analysis

The data confirmed that:

1. The INC_Button is correctly set to the increment function
2. The RES_Button is correctly set to reset the current value to zero
3. The display followed the correct segment configuration defined in the code

Digit	Active Segment
0	a,b,c,e,f,g
1	a,g
2	a,b,d,e,f
3	a,b,d,f,g
4	a,c,d,g
5	b,c,d,f,g
6	b,c,d,e,f,g
7	a,b,g
8	a,b,c,d,e,f,g
9	a,b,c,d,f,g

Result

The experiment successfully shows manual control of 7-segment display using Arduino Uno and push buttons.

1. The 7-segment displays the correct value.
2. The value increases by one when the INC_Button is pressed.
3. The value then successfully reset to zero when RES_Button is pressed.
4. The segments of the display are correctly connected to the Arduino code logic.
5. The display worked correctly without error or flickering.

Discussion

1. Interpretation of Results and Implications

The experimental results confirm the successful integration of a mechatronic system combining hardware (Arduino Uno, 7-segment display, and pushbuttons) with digital logic programming. The primary implication of these results is the verification of state management within a microcontroller environment. By using a counter variable (i) that incremented upon a specific button trigger and reset to zero upon another, the experiment demonstrated how software can maintain "memory" of a system's current state.

2. Discrepancies Between Expected and Observed Outcomes

While the final result was a stable and functional display, there are specific discrepancies between theoretical expectations and experimental observations that are noteworthy:

- Logic Mapping Discrepancy: Theoretically, a digit like '1' is typically formed using segments **b** and **c**. However, the data analysis table indicates that for Digit 1, segments **a** and **g** were active. This suggests a discrepancy in the physical wiring or the internal bitmasking in the code compared to standard 7-segment pinouts.

3. Sources of Error and Improvement

a. Mechanical Button Bounce: Push buttons naturally produce multiple electrical signals once pushed, producing multiple signals. By using a debouncer hardware, this situation can be avoided.

b. Human Reaction Time: Since the reset and increment push button is manually pressed, the pressing speed and duration may vary resulting in inconsistent results or missed counts.

Conclusion

The experiment successfully achieved its primary objective of demonstrating the working principles of digital logic and circuit control through the implementation of a manual counter system. By interfacing an Arduino Uno with a common cathode 7-segment display and tactile pushbuttons, the team verified that numerical values could be precisely controlled and displayed in a sequential manner.

Key findings from the experiment include:

- Functional Integration: The hardware and software were correctly integrated, as evidenced by the 7-segment display accurately representing the values 0 through 9 in response to physical inputs.
- Control Logic Validation: The implementation of the INC_Button for incrementing and the RES_Button for resetting effectively demonstrated the application of conditional logic and state management within a microcontroller environment.

In summary, the successful operation of this setup confirms that the fundamental principles of mechatronic system integration, specifically the relationship between input signals, processing logic, and visual output were applied effectively. This project serves as a foundational step toward understanding more complex digital display systems and user-interface interactions in mechatronic engineering.

Recommendations

1. Enhance User Interface by adding more push buttons for more additional functions such as decrement, pause, or auto-count mode to make the system more interactive.
2. Improve Display Range by using two 7-segment displays to show two-digit numbers (00-99). This can be achieved by using a multiplexer or by connection and additional display with separate control pin.

References

- In depth guide on Seven Segment Display
(https://lastminuteengineers.com/seven-segment-arduino-tutorial)

Acknowledgments

All praise belongs to Allah SWT for granting us the strength and wisdom to complete the "Digital Logic System" experiment. We are grateful for His guidance in navigating the challenges faced throughout the circuit integration process. We also wish to acknowledge the vital support of our lecturers and fellow peers. Their academic supervision and practical insights into mechatronic system integration were essential to the successful operation of our project and the completion of this report.

Student's Declaration

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has not been done by only one individual and all of us have contributed to the report.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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Matric Number: 2314077 Agree [/]

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